

Physics 7B — Final Exam Review Session

Worked Solutions with Teaching Notes

Based on the Spring 2005 Final Exam (J. Liphardt)

UC Berkeley — Spring 2026

How to use this packet. Each problem is followed by a worked solution and a “Teaching Notes” box that highlights the key physics concept, common student mistakes, and conceptual checkpoints worth emphasizing in section. Numerical answers are boxed in green. Compared to the Packard 2001 review packet, this Liphardt 2005 exam emphasizes EM waves, AC circuits / inductor sensors, and statistical-mechanics ideas (equipartition).

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Problem 1 Discharging Capacitor: From Joule Heating to Entropy (18 pts)

Problem 1 (18 pts). A capacitor $C = 3 \mu\text{F}$ is charged with $Q = 4 \times 10^{-4} \text{ C}$ and is connected in series with a wire of length $\ell = 2 \text{ m}$. The wire has a circular cross-section of radius $r = 1 \text{ mm}$, a resistivity of $4 \times 10^{-8} \Omega \text{ m}$, a mass of 10^{-4} kg , and a specific heat of $500 \text{ J}/(\text{kg } ^\circ\text{C})$. At $t = 0$, the capacitor begins to discharge through the resistor. Use $k_B = 1.38 \times 10^{-23} \text{ J/K}$.

- When does the voltage on the capacitor drop to 23 V ?
- When the voltage across the capacitor is 23 V , what is the voltage across the resistor?
- How much thermal energy is deposited in the wire by the discharge process?
- What is the final temperature of the wire? Assume the wire had an initial temperature of 20°C , and no heat is lost to the environment.
- What is the final average kinetic energy of the electrons in the wire?
- What is the change of entropy of the universe in this process? (Conserve energy and don't violate the second law.)

Solution

Setup. Compute the wire's resistance:

$$R = \frac{\rho \ell}{\pi r^2} = \frac{(4 \times 10^{-8})(2)}{\pi(10^{-3})^2} \approx 0.0255 \Omega.$$

The initial capacitor voltage:

$$V_0 = \frac{Q_0}{C} = \frac{4 \times 10^{-4}}{3 \times 10^{-6}} \approx 133.3 \text{ V}.$$

Time constant $RC = (0.0255)(3 \times 10^{-6}) \approx 7.5 \times 10^{-8} \text{ s}$.

Part (a) — Time to drop to 23 V. $V_C(t) = V_0 e^{-t/RC}$. Solving:

$$t = -RC \ln\left(\frac{V_C}{V_0}\right) = -RC \ln\left(\frac{23 \cdot C}{Q_0}\right) = -(7.5 \times 10^{-8}) \ln\left(\frac{23 \cdot 3 \times 10^{-6}}{4 \times 10^{-4}}\right).$$

Numerically, the argument is ≈ 0.1725 , so $\ln(0.1725) \approx -1.76$, giving

$$t \approx (7.5 \times 10^{-8})(1.76) \approx 1.3 \times 10^{-7} \text{ s}.$$

$$t \approx 1.3 \times 10^{-7} \text{ s} = 130 \text{ ns}$$

Part (b) — Voltage across the resistor. By Kirchhoff's voltage law (no other elements in the loop), $V_R(t) = V_C(t)$ *exactly* during the discharge:

$$V_R = 23 \text{ V}.$$

Part (c) — Total thermal energy deposited. By energy conservation, all of the capacitor's initial energy ends up as heat in the wire:

$$U_{\text{thermal}} = \frac{1}{2} \frac{Q_0^2}{C} = \frac{1}{2} \frac{(4 \times 10^{-4})^2}{3 \times 10^{-6}} \approx \boxed{0.027 \text{ J} = 27 \text{ mJ.}}$$

Part (d) — Final temperature of the wire. Apply $Q = mc\Delta T$:

$$\Delta T = \frac{Q}{mc} = \frac{0.027}{(10^{-4})(500)} = 0.54^\circ\text{C} \approx 0.6^\circ\text{C}.$$

So $T_f \approx 20.6^\circ\text{C}$.

$$\boxed{T_f \approx 20.6^\circ\text{C}}$$

Part (e) — Final average KE of the electrons. By equipartition, each translational degree of freedom contributes $\frac{1}{2}k_B T$. The electrons in the wire form (approximately) a 3-D ideal gas, so:

$$\langle KE \rangle = \frac{3}{2} k_B T_f = \frac{3}{2} (1.38 \times 10^{-23}) (20.6 + 273.15) \approx \boxed{6.08 \times 10^{-21} \text{ J.}}$$

Part (f) — Entropy change of the universe. The entropy change of the wire as it heats from T_i to T_f at constant pressure is

$$\Delta S_{\text{wire}} = \int_{T_i}^{T_f} \frac{dQ}{T} = \int_{T_i}^{T_f} \frac{mc dT}{T} = mc \ln \frac{T_f}{T_i}.$$

Plugging in (in Kelvin: $T_i = 293.15 \text{ K}$, $T_f = 293.69 \text{ K}$):

$$\Delta S = (10^{-4})(500) \ln \frac{293.69}{293.15} \approx (0.05)(1.84 \times 10^{-3}) \approx \boxed{1.02 \times 10^{-4} \text{ J/K.}}$$

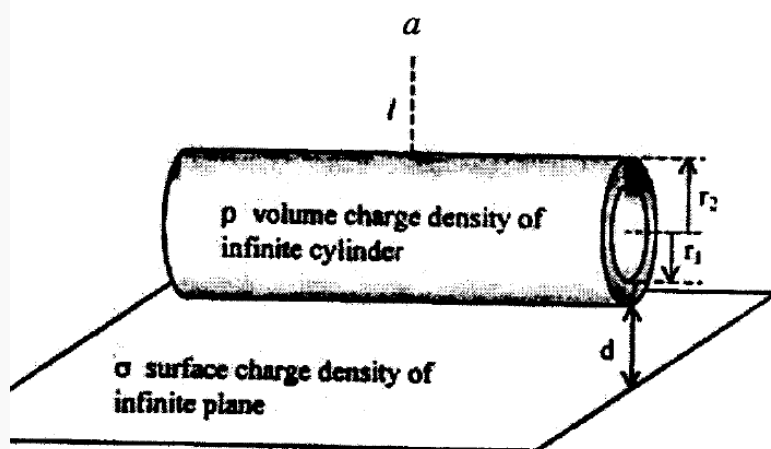
The capacitor goes to a definite state ($Q = 0$), so its entropy is unchanged. The universe's $\Delta S > 0$, as required.

Teaching Notes & Common Pitfalls

- **Six parts, three real ideas.** (a)–(c) are the standard RC discharge + energy-conservation chain. (d)–(f) bridge to thermal physics — this is what makes Liphardt's problem distinctive.
- **Part (e) is sneaky.** It asks about *electrons in the wire*, not the bulk wire. Use equipartition: $\langle KE \rangle = \frac{3}{2} k_B T$. The 0.6°C heating from part (d) feeds directly into the electron energy.
- **ΔS for irreversible processes.** Even though the discharge is irreversible, ΔS for the wire is computed via a *reversible path* between the same endpoints. That's why we integrate $mc dT/T$ as if heating slowly.

Problem 2 Charged Particle Falling onto a Cylinder + Sheet (20 pts)

Problem 2 (20 pts). An infinitely long, non-conducting cylinder has thickness $r_2 - r_1$. The cylinder has a uniform volume charge density ρ and is positioned a distance d away from an infinite non-conducting sheet with surface charge density σ . A charged particle (mass m and charge $-q$) is released from point a , at a height ℓ above the cylinder. What is the kinetic energy of the particle when it collides with the cylinder? Neglect gravity.



Solution

Setup. Two field sources, both pointing upward at point a (toward the falling charge):

- Infinite sheet: $E_{\text{sheet}} = \sigma/(2\epsilon_0)$ (uniform, perpendicular to sheet).
- Cylinder (using a Gaussian cylinder of radius $r > r_2$):

$$E(r) \cdot 2\pi r \ell_G = \frac{\rho \cdot \pi(r_2^2 - r_1^2) \ell_G}{\epsilon_0} \implies E_{\text{cyl}}(r) = \frac{\rho(r_2^2 - r_1^2)}{2\epsilon_0 r}.$$

The negative charge feels a force *into* the field direction (i.e., *toward* the cylinder), so it accelerates downward.

Energy via work integral. The particle starts at distance $r_2 + \ell$ from the cylinder's axis and stops at distance r_2 (its surface). The total field along its path is

$$E_{\text{tot}}(r) = \frac{\sigma}{2\epsilon_0} + \frac{\rho(r_2^2 - r_1^2)}{2\epsilon_0 r}.$$

The work done on the charge (which has charge $-q$ and moves in the direction *opposite* to $\vec{E}$) is:

$$W = \int_{r_2+\ell}^{r_2} \vec{F} \cdot d\vec{r} = (-q) \int_{r_2+\ell}^{r_2} E_{\text{tot}}(-1) dr = q \int_{r_2}^{r_2+\ell} E_{\text{tot}}(r) dr.$$

(The factor of -1 inside the integral comes from $d\vec{r}$ pointing inward while $\vec{E}$ points outward.)
Carry out the integral:

$$W = q \int_{r_2}^{r_2+\ell} \left[\frac{\sigma}{2\epsilon_0} + \frac{\rho(r_2^2 - r_1^2)}{2\epsilon_0 r} \right] dr = \frac{q\sigma\ell}{2\epsilon_0} + \frac{q\rho(r_2^2 - r_1^2)}{2\epsilon_0} \ln \frac{r_2 + \ell}{r_2}.$$

By the work-energy theorem, $K = W$:

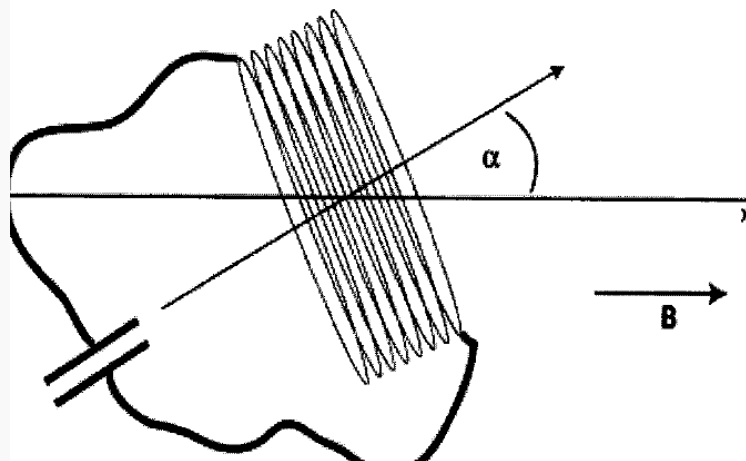
$$K = \frac{q}{2\epsilon_0} \left[\sigma\ell + \rho(r_2^2 - r_1^2) \ln \frac{r_2 + \ell}{r_2} \right].$$

Teaching Notes & Common Pitfalls

- **Why $r_2^2 - r_1^2$, not r_2^2 ?** The cylinder is hollow (a thick shell). Only the charge between r_1 and r_2 is enclosed, contributing area $\pi(r_2^2 - r_1^2)$ in cross-section.
- **The “superposition is your friend” lesson.** Treat the sheet and cylinder as two independent sources. Compute each field separately, then add. *Then* integrate to get the work.
- **Sign vigilance.** The charge is $-q$, the field points outward from the cylinder, but the charge moves inward. Two minus signs cancel and the work is positive (KE increases) — as it should, since the negative charge is attracted toward the positive cylinder. Encourage students to verify the sign by physical intuition before plowing through the integral.
- **Why no gravity?** The problem says “neglect gravity.” Make sure students don’t sneak in $mg\ell$ terms.
- **What about the sheet’s contribution depending on side?** The infinite-sheet field is uniform and points away from the sheet on both sides. Since the cylinder sits between the sheet and point a , we need the sheet’s field on the side of the cylinder *above* where the charge falls — it points up (away from the sheet), and this contributes to attracting the negative charge downward.

Problem 3 Magnetic Flux Sensor: Inductor + Capacitor (20 pts)

Problem 3 (20 pts) — Magnetic flux detection. You're an engineer trying to design a magnetic flux sensor. You decide to connect an inductor (inductance L) in parallel with a capacitor (capacitance C), yielding a resonant circuit with frequency ω . The external magnetic field points in the x direction.



- Consider just the inductor. Assume it has a **constant** angle α relative to $\vec{B}$. The inductor has N turns, area A , length ℓ , and carries current I . What is the total voltage induced between the two leads, as a function of dI/dt , dB/dt , and α ?
- Why won't the sensor in (a) work for *static* $\vec{B}$ (i.e. $dB/dt = 0$)? Answer in one sentence.
- Improve the sensor by spinning the inductor: $\alpha(t) = 2\pi\nu t$. Now what is the induced voltage as a function of dI/dt , dB/dt , and ν ?
- Set up (but do not solve) the differential equation for the full sensor (inductor + capacitor) using Kirchhoff's loop rule.

Solution

Part (a) — Two contributions to the flux. Total flux through the inductor has two pieces:

1. **From external $\vec{B}$:** $\Phi_{\text{ext}} = NBA \cos \alpha$.

2. **From self-induced field $B_s = \mu_0 n I$** (where $n = N/\ell$ is turn density):

$$\Phi_{\text{self}} = NB_s A = \frac{\mu_0 N^2 I A}{\ell}.$$

So the total flux is

$$\Phi_B = NBA \cos \alpha + \frac{\mu_0 N^2 I A}{\ell}.$$

By Faraday's law, $\mathcal{E} = -d\Phi_B/dt$. With α constant:

$$|\mathcal{E}| = NA \cos \alpha \frac{dB}{dt} + \frac{\mu_0 N^2 A}{\ell} \frac{dI}{dt}.$$

The second coefficient is precisely the inductance: $L = \mu_0 N^2 A/\ell$.

Part (b) — Why static fields don't register.

The induced voltage depends on dB/dt and dI/dt , neither of which depends on B itself when $dB/dt = 0$. So the sensor produces no signal that contains information about a static B .

Part (c) — Spinning the inductor. Now $\alpha(t) = 2\pi\nu t$, so we must use the product rule on $\cos \alpha$:

$$\frac{d\Phi_B}{dt} = NA \cos \alpha \frac{dB}{dt} - NAB \sin \alpha \frac{d\alpha}{dt} + \frac{\mu_0 N^2 A}{\ell} \frac{dI}{dt}.$$

With $d\alpha/dt = 2\pi\nu$:

$$|\mathcal{E}| = NA \cos(2\pi\nu t) \frac{dB}{dt} - 2\pi\nu NAB \sin(2\pi\nu t) + \frac{\mu_0 N^2 A}{\ell} \frac{dI}{dt}.$$

The middle term is the new term from rotation — it survives even when $dB/dt = 0$, so the sensor now responds to static fields.

Part (d) — The differential equation. Going around the loop and summing potential drops to zero:

$$\mathcal{E} + \frac{Q}{C} = 0,$$

where Q is the charge on the capacitor and $I = dQ/dt$. Substituting from part (c) and rewriting $dI/dt = d^2Q/dt^2$:

$$\frac{\mu_0 N^2 A}{\ell} \frac{d^2 Q}{dt^2} + \frac{Q}{C} = 2\pi\nu NAB \sin(2\pi\nu t) - NA \cos(2\pi\nu t) \frac{dB}{dt}.$$

This is the driven-LC oscillator equation, with the right-hand side as the time-varying flux drive.

Teaching Notes & Common Pitfalls

- **Big idea: two ways flux can change** — the field can vary, or the coil's orientation can vary. Most problems give you only one; this one deliberately gives both.
- **Why L shows up automatically.** The coefficient $\mu_0 N^2 A/\ell$ on dI/dt is the textbook self-inductance of a solenoid. Liphardt shows where L comes from, not just $V = L dI/dt$.
- **Mechanical analogy for (b)/(c).** A static B -field is like a static gravitational field: a stationary EM sensor can't detect it. You must move (rotate) to "feel" it.
- **Practical application:** rotating-coil sensors are exactly how fluxgate magnetometers work, and historically how Earth's static field was measured.

Problem 4 An EM Wave: Direction, Magnitude, Wavelength, Power (15 pts)

Problem 4 (15 pts) — Light. Let there be light:

$$\vec{E} = 4\hat{z} \sin\left(\frac{2\pi}{\lambda}x + 5\pi \times 10^{15}t\right) \text{ V/m}$$

$$\vec{B} = \vec{B}_0 \sin\left(\frac{2\pi}{\lambda}x + 5\pi \times 10^{15}t\right) \text{ T}$$

- (a) In what direction does the wave propagate?
- (b) What is $\vec{B}_0$ (magnitude and direction) if the wave is in vacuum?
- (c) How does (b) change in water ($\varepsilon = 1.79\varepsilon_0$, $\mu = \mu_0$)?
- (d) Wavelength in vacuum and in water?
- (e) Maximum value of $|\vec{S}|$ (Poynting vector) in vacuum and water?

Solution

Part (a) — Direction of propagation. The phase has the form $kx + \omega t$ with both terms positive (same sign). A wave moving in the $+\hat{n}$ direction has phase $kx - \omega t$; the opposite sign means the wave moves in the $-\hat{x}$ direction:

Propagates in the $-\hat{x}$ direction.

Part (b) — $\vec{B}_0$ in vacuum. The relation $|\vec{B}| = |\vec{E}|/c$ (in any EM wave in vacuum) gives:

$$B_0 = \frac{E_0}{c} = \frac{4}{3 \times 10^8} \approx 1.3 \times 10^{-8} \text{ T.}$$

Direction: $\vec{E} \times \vec{B}$ must point in the propagation direction $-\hat{x}$. With $\vec{E}$ in $+\hat{z}$, we need $\hat{z} \times \hat{B} = -\hat{x}$, so $\hat{B} = +\hat{y}$:

$$\vec{B}_0 = (1.3 \times 10^{-8} \text{ T})\hat{y}$$

Part (c) — In water. In a medium, the speed is $v = 1/\sqrt{\mu\varepsilon}$, so the relation $B_0 = E_0/v$ becomes

$$B_0 = E_0\sqrt{\mu\varepsilon} = E_0\sqrt{\mu_0 \cdot 1.79\varepsilon_0} = \frac{E_0}{c}\sqrt{1.79} \approx (1.3 \times 10^{-8})\sqrt{1.79}.$$

$$B_0 \approx 1.7 \times 10^{-8} \text{ T}$$

Part (d) — Wavelengths. Read ω from the phase: $\omega = 5\pi \times 10^{15} \text{ 1/s}$, so $f = \omega/(2\pi) = 2.5 \times 10^{15} \text{ Hz}$ (UV light).

Vacuum:

$$\lambda_{\text{vac}} = \frac{c}{f} = \frac{3 \times 10^8}{2.5 \times 10^{15}} \approx 1.2 \times 10^{-7} \text{ m} = 120 \text{ nm}.$$

Water: The frequency stays the same; the speed drops by $\sqrt{1.79}$:

$$\lambda_{\text{water}} = \frac{\lambda_{\text{vac}}}{\sqrt{1.79}} \approx \frac{1.2 \times 10^{-7}}{1.338} \approx 9.0 \times 10^{-8} \text{ m} = 90 \text{ nm}.$$

$$\lambda_{\text{vac}} \approx 120 \text{ nm}, \quad \lambda_{\text{water}} \approx 90 \text{ nm}$$

Part (e) — Maximum Poynting magnitude. Generally $\vec{S} = \frac{1}{\mu} \vec{E} \times \vec{B}$. For an EM wave with $\vec{E} \perp \vec{B}$:

$$S_{\text{max}} = \frac{E_0 B_0}{\mu}.$$

Vacuum:

$$S_{\text{max,vac}} = \frac{(4)(1.3 \times 10^{-8})}{4\pi \times 10^{-7}} \approx 0.04 \text{ W/m}^2.$$

Water (still $\mu = \mu_0$, but B_0 is bigger):

$$S_{\text{max,water}} = \frac{(4)(1.7 \times 10^{-8})}{4\pi \times 10^{-7}} \approx 0.05 \text{ W/m}^2.$$

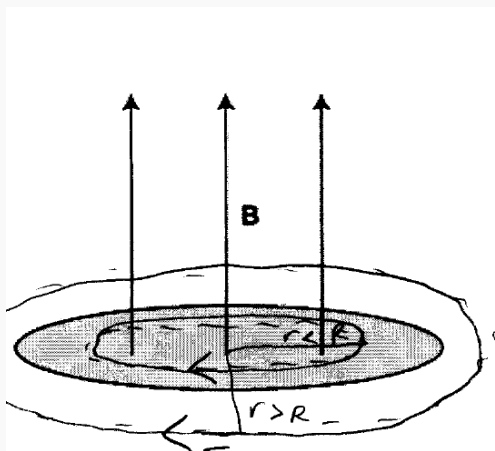
$$S_{\text{max,vac}} \approx 0.04 \text{ W/m}^2, \quad S_{\text{max,water}} \approx 0.05 \text{ W/m}^2$$

Teaching Notes & Common Pitfalls

- **Sign convention for direction.** The recipe: in $\sin(kx \pm \omega t)$, the wave moves in the direction that makes the argument constant as t increases. With $+\omega t$, x must *decrease*, so the wave moves in $-\hat{x}$. Drill this with students — it's a perpetual source of mistakes.
- **Why $\vec{E} \times \vec{B}$ tells you direction.** The Poynting vector points in the direction of energy flow, which for a plane wave is the propagation direction. Use this as a cross-check rather than trying to memorize it.
- **Frequency invariance at boundaries.** When light crosses from vacuum to water, the *frequency* stays fixed (driven by the source) but the wavelength shrinks because the speed drops. This is why the index of refraction $n = \sqrt{\epsilon_r}$ scales λ but not f .
- **Numerical sanity:** 120 nm is far UV — this isn't visible light. Students sometimes panic when the answer doesn't match their intuition for "light."
- **Watch the Poynting formula.** For a non-magnetic medium ($\mu = \mu_0$), use $S = EB/\mu_0$. Don't accidentally write $S = EB/\mu_{\text{water}}$ and screw up the water case — water is non-magnetic, so μ_0 is correct.

Problem 5 Faraday's Law: Induced E-Field Inside and Outside a Region (10 pts)

Problem 5 (10 pts) — Faraday's law. A uniform magnetic field $\vec{B}$, pointing straight up, fills the circular region with radius R . If $\vec{B}$ is changing with time, what is the induced electric field $\vec{E}$ in the plane of the circle for $r < R$, and $r > R$?



Solution

Setup. Use a circular Ampèrian-style loop of radius r in the plane perpendicular to $\vec{B}$. By the symmetry, $\vec{E}$ is azimuthal and depends only on r :

$$\oint \vec{E} \cdot d\vec{\ell} = E(r) \cdot 2\pi r.$$

Faraday's law:

$$\oint \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}.$$

Region 1: $r < R$. The flux enclosed is $\Phi_B = B \cdot \pi r^2$ (the full loop is inside the B region):

$$E(r) \cdot 2\pi r = -\pi r^2 \frac{dB}{dt} \implies \boxed{E(r) = \frac{r}{2} \frac{dB}{dt} \quad (r < R)}$$

By Lenz's law, if B points up and is increasing, the induced E drives current clockwise as viewed from above (i.e., the induced current would oppose the increase by creating downward flux).

Region 2: $r > R$. Now the enclosed flux saturates at $\Phi_B = B \cdot \pi R^2$ (only the central region of radius R has $B \neq 0$):

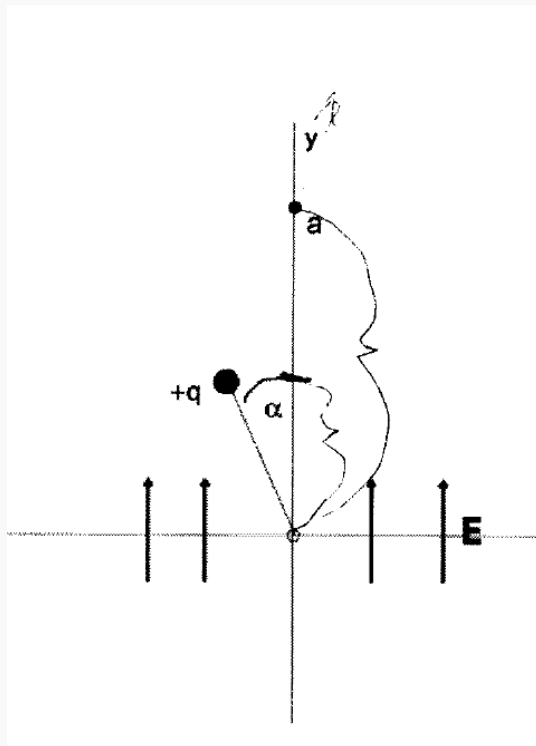
$$E(r) \cdot 2\pi r = -\pi R^2 \frac{dB}{dt} \implies \boxed{E(r) = \frac{R^2}{2r} \frac{dB}{dt} \quad (r > R)}$$

Teaching Notes & Common Pitfalls

- **This is the magnetic dual of Gauss's law for charged cylinders.** Inside, $E \propto r$; outside, $E \propto 1/r$. Pointing this duality out helps students remember both regions in one shot.
- **Why $E \propto 1/r$ outside?** The total “cause” of the induced field — the changing flux — is fixed at $\pi R^2(dB/dt)$ for $r > R$. Spreading that “cause” over a longer Ampèrian loop ($2\pi r$) gives $E \propto 1/r$. Same reason the magnetic field of a wire goes as $1/r$.
- **Lenz's law direction.** Always: “the induced field opposes the change in flux.” Don't get tangled in $-d\Phi/dt$ signs — physically reason about which direction of $\vec{E}$ would drive a current that creates flux opposing dB/dt .
- **Continuity at $r = R$.** Plug in $r = R$ to both expressions: they agree, $E(R) = R/2 \cdot dB/dt$. This is a great consistency check students can do automatically.

Problem 6 Charge on a Rod in an External Field — and Its Thermal Fluctuations (25 pts)

Problem 6 (25 pts) — Charge fluctuations. Consider an electric charge $+q$ in an external field $\vec{E}$. The charge is connected to the origin with a rod of length ℓ .



- Lowest-energy configuration?
- Highest-energy configuration?
- At which angle is the torque largest?
- Total electric field at point a (height h), as a function of α .
- Find $U(\alpha)$, electrostatic potential energy, with $U(0) = 0$. [Hint: for small α , $\cos \alpha \approx 1 - \alpha^2/2$.]
- Use (e) to find the magnitude of the charge's thermal fluctuations at temperature T . (If you couldn't do (e), assume $U(\alpha) = qE\alpha^2\ell$.)
- Does the capacitance of a capacitor increase with temperature? Yes / No.

Solution

Parts (a)–(c) — Quick angles. The dipole-like moment $q\ell$ tries to align with $\vec{E}$ (the same way a bar magnet aligns with $\vec{B}$):

- (a) $\alpha = 0$: the rod points up along $\vec{E}$.

$$\alpha_{\min} = 0^\circ.$$

- (b) $\alpha = 180^\circ$: the rod points opposite to $\vec{E}$ (unstable).

$$\alpha_{\max} = 180^\circ.$$

- (c) Torque magnitude is $|\vec{\tau}| = qE\ell \sin \alpha$, maximum at:

$$\alpha = 90^\circ.$$

Part (d) — Total $\vec{E}$ at point a . Two contributions at point $a = (0, h)$:

1. External field: $\vec{E}_{\text{ext}} = E\hat{y}$.
2. Field from the point charge $+q$ at position $(-\ell \sin \alpha, \ell \cos \alpha)$.

The vector from $+q$ to point a has components

$$\Delta x = +\ell \sin \alpha, \quad \Delta y = h - \ell \cos \alpha.$$

The distance squared is

$$d^2 = (\ell \sin \alpha)^2 + (h - \ell \cos \alpha)^2 = h^2 + \ell^2 - 2h\ell \cos \alpha$$

(by the law of cosines, exactly as the figure indicates).

The field from $+q$ at point a is

$$\vec{E}_q = \frac{q}{4\pi\epsilon_0 d^3} (\Delta x \hat{x} + \Delta y \hat{y}),$$

so

$$\vec{E}_{\text{tot}} = \frac{q \ell \sin \alpha}{4\pi\epsilon_0 (h^2 + \ell^2 - 2h\ell \cos \alpha)^{3/2}} \hat{x} + \left[E + \frac{q(h - \ell \cos \alpha)}{4\pi\epsilon_0 (h^2 + \ell^2 - 2h\ell \cos \alpha)^{3/2}} \right] \hat{y}.$$

Part (e) — Potential energy $U(\alpha)$. The work needed to rotate the dipole from $\alpha = 0$ to angle α in a uniform field is

$$U(\alpha) = -\vec{p} \cdot \vec{E} + \text{const} = -qE\ell \cos \alpha + C_0.$$

Setting $U(0) = 0$: $0 = -qE\ell + C_0$, so $C_0 = qE\ell$, and

$$U(\alpha) = qE\ell(1 - \cos \alpha).$$

Small-angle approximation, $\cos \alpha \approx 1 - \alpha^2/2$:

$$U(\alpha) \approx \frac{1}{2}qE\ell \alpha^2$$

This is harmonic (quadratic in α) — a textbook 1-D harmonic oscillator with effective spring constant $\kappa_{\text{eff}} = qE\ell$.

Part (f) — Thermal fluctuations via equipartition. The system has **one** degree of freedom (the angle α). Equipartition assigns $\frac{1}{2}k_B T$ of energy to each quadratic degree of freedom in the Hamiltonian:

$$\langle U \rangle = \frac{1}{2}k_B T \implies \frac{1}{2}qE\ell \langle \alpha^2 \rangle = \frac{1}{2}k_B T.$$

So the typical (RMS) thermal fluctuation in α is

$$\alpha_{\text{rms}} = \sqrt{\frac{k_B T}{qE\ell}}.$$

Part (g) — Capacitance vs. temperature. Yes. Heating a dielectric increases the average dipole alignment fluctuations (or in the same vein, can increase the dielectric's polarizability), so the dielectric constant ϵ tends to grow modestly with T for many materials, increasing $C = \epsilon A/d$. (Caveat: for some materials ϵ *decreases* with T , so the answer is more nuanced in real life. The expected answer here is “Yes,” tied to part (f).)

Teaching Notes & Common Pitfalls

- **The hidden message of this whole problem.** Liphardt is building students up to a very deep idea: *a dipole in a field is just a harmonic oscillator*, and any harmonic oscillator at temperature T has thermal fluctuations of size $\sqrt{k_B T / \kappa}$. This is the same logic behind the Johnson noise in resistors, the phonon zero-point in crystals, and Brownian motion. Worth flagging as a unifying lens.
- **Equipartition counts *quadratic* degrees of freedom.** Each $\frac{1}{2}m\dot{x}^2$ and each $\frac{1}{2}\kappa x^2$ is one DOF and gets $\frac{1}{2}k_B T$. Here we focused only on the potential side (giving fluctuations); a full treatment would also assign $\frac{1}{2}k_B T$ to the kinetic side, $\frac{1}{2}I\dot{\alpha}^2$.
- **The trick in part (e):** the integration constant is fixed by the boundary condition $U(0) = 0$, which is the “standard” way physicists choose the zero of potential energy. Don't let students leave $-qE\ell \cos \alpha$ as the answer if the problem specifies $U(0) = 0$.
- **Part (d) doesn't actually feed into anything else.** It's a self-contained “can you do vector addition + law of cosines” check. Don't get bogged down in algebra during the review — just demonstrate the setup and skip to the result.
- **Sanity check on (f):** for “everyday” values — q = an electron's charge, $E = 1 \times 10^6$ V/m (a strong lab field), ℓ = one nanometer, T = room temperature — α_{rms} is on the order of 1 radian. Thermal fluctuations matter when $k_B T \sim qE\ell$, which is a great way to think about whether dielectric polarization will be ordered or thermally scrambled.

End of review packet — good luck on the final!